International Journal of Biological Macromolecules 165 (2020) 187–197

Contents lists available at ScienceDirect

International Journal of Biological Macromolecules

journal homepage: http://www.elsevier.com/locate/ijbiomac

# Effect of hydrogen bond donor on the choline chloride-based deep eutectic solvent-mediated extraction of lignin from pine wood

Yujin Oh [a,1], Saerom Park [a,1], Dahun Jung [a], Kyeong Keun Oh [b], Sang Hyun Lee [a,*]

[a] Department of Biological Engineering, Konkuk University, Seoul 05029, South Korea
[b] Department of Chemical Engineering, Dankook University, Yongin 16890, Gyeonggi, South Korea

ARTICLE INFO

*Article history:*
Received 21 July 2020
Received in revised form 16 September 2020
Accepted 19 September 2020
Available online 28 September 2020

*Keywords:*
Deep eutectic solvents
Lignocellulose
Solvatochromic parameters

ABSTRACT

In this work, twenty-five kinds of choline chloride (ChCl)-based deep eutectic solvents (DESs) containing acid, hydroxyl, amide, and binary hydrogen bond donors (HBDs) were prepared and successfully used to pretreat pine wood powder. As a result of the pretreatment, the glucan content in the pretreated biomass was increased, whereas the contents of hemicellulose and lignin were significantly decreased. The biomass pretreatment efficiency of the DESs had improved with increasing the polarity and hydrogen bond acidity ($\alpha$) of the DESs. Among the studied DESs, ChCl:lactic acid:formic acid (1:1:1) with the highest $\alpha$ value was the most efficient DES in extracting lignin from biomass. The pretreated biomass also showed an enhanced enzymatic saccharification yield owing to the decreased particle size of the biomass and reduced content of hemicellulose and lignin. During the pretreatment process of biomass using DESs, the extracted lignin could be recovered successfully, with a yield of up to 60% and purity of over 90%. The molecular weight of the extracted lignin was much lower than that of the native cellulolytic enzyme lignin. The DES used for pretreatment process could be also successfully reused with high recovery yield of DES and high retention of delignification capacity.

© 2020 Elsevier B.V. All rights reserved.

## 1. Introduction

Lignocellulosic biomass, such as agricultural residue, forestry waste, and waste paper, is a considerably abundant resource on earth and has long been considered as a potential sustainable source for biotransformation into value-added bioproducts. However, the utility of lignocellulosic biomass has been limited due to its recalcitrant structure [1–3]. The three primary biopolymers of lignocellulose are cellulose, hemicellulose, and lignin. Cellulose is highly crystalline, and the enzymatic hydrolysis of cellulose is hindered by the existence of hemicellulose and lignin. Hemicellulose forms hydrogen bonds with cellulose and is covalently linked with lignin. Lignin is a highly branched aromatic polymer and acts as the glue that binds cellulose and hemicellulose in the lignocellulosic biomass. To improve the usability of lignocellulose by enhancing the accessibility of enzymes, most pretreatment strategies have focused on removing the lignin from the lignocellulose and reducing the crystallinity [4]. Various pretreatment methods, such as steam or ammonia explosion, acid hydrolysis, alkaline hydrolysis, high-temperature organic solvent pretreatment, and enzymatic degradation, have been developed [5–7]. However, recent research has focused on the development of more energy-efficient and environmentally friendly processes to selectively remove the lignin from lignocellulose and reduce the degradation of cellulose.

Recently, deep eutectic solvents (DESs) have been applied as green solvents for the pretreatment of lignocellulosic biomass. DESs have emerged as alternatives to organic solvents and ionic liquids (ILs) owing to their nonvolatility, thermal stability, and nontoxicity. DESs can be prepared by simply mixing a hydrogen bond acceptor (HBA) with one or two hydrogen bond donors (HBD) and do not require purification [8–10]. Choline chloride (ChCl) is the most frequently used HBA in the preparation of DESs because of its inexpensive price, biocompatibility, and biodegradability [11]. Various HBDs, such as alcohols, amides, acids, and polyols, can be used to prepare the DESs. The properties of DESs can be easily controlled by changing the type and molar ratio of the HBA and HBD [12]. Therefore, biomass pretreatment processes using DESs have attracted attention in recent years. Alvarez-Vasco et al. reported that DESs are promising for the extraction of low molecular weight lignin from Douglas firs and poplar [13]. Xing et al. showed that dihydrogen-bonding DESs can be used as reaction media for the pretreatment of rice straw to enhance the fermentation yield [14]. Sai and Lee verified that ChCl:lactic acid is a promising pretreatment solvent for empty fruit bunches [15]. Liu et al. demonstrated that ChCl:lactic acid can efficiently delignify and produce nanocellulose from moso bamboo [16]. Liu et al. applied a microwave-assisted treatment using DESs to cleave the lignin-carbohydrate complex and extract lignin oligomers from poplar wood [1]. However, because the

* Corresponding author.
*E-mail address:* sanghlee@konkuk.ac.kr (S.H. Lee).
[1] These authors contributed equally.

https://doi.org/10.1016/j.ijbiomac.2020.09.145
0141-8130/© 2020 Elsevier B.V. All rights reserved.

physicochemical properties of DESs have not been clearly elucidated, previous studies have consisted primarily of arbitrarily screening suitable DESs. Therefore, it is necessary to clarify the physicochemical properties of DESs and systematically analyze their effects during biomass pretreatment.

Solvatochromic parameters are especially useful to understand the physicochemical characteristics of solvents and are substantially correlated with various solvent properties such as extraction rate, chemical reaction rate, and solubility. These parameters can be easily measured using dyes with various UV–visible spectra depending on the nature of the solvent [17,18]. The most widely used solvatochromic parameters are the polarity scale ($E_T$) and the Kamlet-Taft parameters, including dipolarity/polarizability ($\pi^*$), hydrogen bond acidity ($\alpha$), and hydrogen bond basicity ($\beta$). The betaine derivative $E_T$ (30), also known as Reichardt's dye, is the most widely used dye to determine the $E_T$ scale. However, because Reichardt's dye is sensitive to acid, dichloro-substituted betaine $E_T$(33) and Nile red are more suitable for measuring the polarity of acidic solvents, such as DESs containing acid HBDs [19]. The solvatochromic parameters of green solvents, such as ionic liquid (ILs) and DESs, have been useful in understanding equilibrium and kinetic phenomena recently [8,20,21]. For example, Doherty el al. reported that the solubility of biomass in ILs correlated well with the $\beta$ values of the ILs [22].

In this study, twenty-five types of ChCl-based DESs containing one or two HBDs were prepared and used as pretreatment solvents for pine wood powder. The effects of HBDs on the pretreatment efficiency for such factors as the recovery of solid residue after treatment, hemicellulose removal, lignin removal, saccharification yield of the pretreated biomass, and recovery of the extracted lignin was investigated. Solvatochromic parameters, including polarity, dipolarity/polarizability, hydrogen bond acidity, and hydrogen bond basicity, were measured to understand the effects of HBDs on the DES properties. The correlation between the solvent properties and the pretreatment efficiency was also studied to explain the pretreatment mechanisms of the DESs on lignocellulosic biomass.

## 2. Materials and methods

### 2.1. Materials

*N*,*N*-Diethyl-4-nitroaniline (**1**) was provided by Frinton Laboratories Inc. (Hainesport, USA). Nile red (**2**), 4-nitroaniline (**3**), choline chloride (ChCl), urea (U), formamide (F), acetamide (A), glycerol (G), ethylene glycol (EG), diethylene glycol (DG), and triethylene glycol (TG) were purchased from Sigma-Aldrich (St. Louis, USA). Acetic acid (AA), citric acid (CA), formic acid (FA), lactic acid (LA), malic acid (MA), propionic acid (PA), succinic acid (SA), and xylitol (X) were supplied by Samchun Pure Chem. Co., Ltd. (Seoul, South Korea). Wood powder from *Pinus rigida* (average particle size: 141 μm) was purchased from G-Biotech (Seoul, South Korea). Celluclast® (63 FPU/mL, cellulase from *Trichoderma reesei*) was obtained from Novozymes (Bagsværd, Denmark). All other chemicals used in this study were of analytical grade and were used without further purification.

### 2.2. Preparation of DESs

Twenty-five DESs were prepared by mixing ChCl with various HBD partners at a specific molar ratio (Table 1). The mixtures were stirred at 80 °C for 2 h and then left to cool to room temperature [23]. All prepared DESs were liquid at room temperature except ChCl:A.

### 2.3. Pretreatment of lignocellulosic biomass, recovery of lignin, and recycling of DES

For the pretreatment of lignocellulosic biomass to extract lignin, 1.5 g pine wood powder was mixed with 15 g DES and then stirred at 130 °C for 6 h. After the lignin extraction process, the mixture was cooled at room temperature, and 30 mL ethanol was then added to the mixture to decrease the viscosity of the mixture. The solid and soluble fractions of the mixture were separated via filtration using a glass crucible. The solid fraction (pretreated wood powder) was washed three times with ethanol and then vacuum dried.

**Table 1**
Mixing ratios of prepared DESs.

| DESs | | Molar ratio | Abbreviation | No. |
|---|---|---|---|---|
| Acid HBDs | ChCl:formic acid | 1:2 | ChCl:FA | 1 |
| | ChCl:acetic acid | 1:2 | ChCl:AA | 2 |
| | ChCl:propionic acid | 1:2 | ChCl:PA | 3 |
| | ChCl:lactic acid | 1:2 | ChCl:LA | 4 |
| | ChCl:citric acid | 1:1 | ChCl:CA | 5 |
| | ChCl:malic acid | 1:1 | ChCl:MA | 6 |
| | ChCl:succinic acid | 1:1 | ChCl:SA | 7 |
| Hydroxyl HBDs | ChCl:glycerol | 1:2 | ChCl:G | 8 |
| | ChCl:ethylene glycol | 1:2 | ChCl:EG | 9 |
| | ChCl:diethylene glycol | 1:2 | ChCl:DG | 10 |
| | ChCl:triethylene glycol | 1:2 | ChCl:TG | 11 |
| | ChCl:xylitol | 1:1 | ChCl:X | 12 |
| Amide HBDs | ChCl:urea | 1:2 | ChCl:U | 13 |
| | ChCl:formamide | 1:2 | ChCl:F | 14 |
| | ChCl:acetamide | 1:2 | ChCl:A | 15 |
| Binary HBDs | ChCl:lactic acid:formic acid | 1:1:1 | ChCl:LA:FA | 16 |
| | ChCl:lactic acid:formamide | 1:1:1 | ChCl:LA:F | 17 |
| | ChCl:formic acid:ethylene glycol | 1:0.4:1.6 | $ChCl_1$:$FA_{0.4}$:$EG_{1.6}$ | 18 |
| | ChCl:acetic acid:ethylene glycol | 1:0.4:1.6 | $ChCl_1$:$AA_{0.4}$:$EG_{1.6}$ | 19 |
| | ChCl:propionic acid:ethylene glycol | 1:0.4:1.6 | $ChCl_1$:$PA_{0.4}$:$EG_{1.6}$ | 20 |
| | ChCl:lactic acid:ethylene glycol | 1:0.4:1.6 | $ChCl_1$:$LA_{0.4}$:$EG_{1.6}$ | 21 |
| | ChCl:formic acid:ethylene glycol | 1:1.2:0.8 | $ChCl_1$:$FA_{1.2}$:$EG_{0.8}$ | 22 |
| | ChCl:acetic acid:ethylene glycol | 1:1.2:0.8 | $ChCl_1$:$AA_{1.2}$:$EG_{0.8}$ | 23 |
| | ChCl:propionic acid:ethylene glycol | 1:1.2:0.8 | $ChCl_1$:$PA_{1.2}$:$EG_{0.8}$ | 24 |
| | ChCl:lactic acid:ethylene glycol | 1:1.2:0.8 | $ChCl_1$:$LA_{1.2}$:$EG_{0.8}$ | 25 |

For the lignin recovery, the soluble fraction containing the extracted lignin was evaporated to eliminate the ethanol, and then 100 mL distilled water was added to precipitate the lignin at 4 °C. The precipitated lignin was washed three times with a water/ethanol (9:1) mixture and then vacuum dried. The lignin yield was calculated using the following equation:

$$\text{Lignin yield}\ (\%) = \frac{\text{Mass of the recovered lignin}}{\text{Mass of the lignin in the initial sample}} \times 100 \quad (1)$$

For the recycling of DES used for pretreatment of biomass, the mixture of DES and water after lignin precipitation was vacuum evaporated to remove water and then additionally dried at 105 °C for 12 h. Recovered DES was reused without further purification to pretreat biomass by the same procedure described above. The DES was reused five times.

### 2.4. Compositional analysis of pretreated biomass

The chemical composition of the pretreated wood powder and recovered lignin was analyzed according to the NREL method [24]. Briefly, the sample was treated with 72% (v/v) sulfuric acid at 30 °C for 2 h, followed by dilute acid (4%) at 121 °C for 1 h. The hydrolysis products (glucose, xylose, galactose, arabinose, and mannose) were determined using an HPLC (Young-Lin Model YL9100, South Korea) equipped with an RI detector and a Shodex SP0810 column operated at 85 °C. The mobile phase consisted of deionized water with a flow rate of 0.6 mL/min.

The glucan content was calculated by multiplying the conversion factor (considering the release of water molecules after polymerization of glucose) of 0.9 to the total glucose content in the sample. The amount of hemicellulose was calculated as the total amount of xylose, galactose, arabinose, and mannose multiplied by the conversion factor of 0.88 for C-5 and C-6 sugars. The filtered solid residue after acid hydrolysis was dried at 105 °C and was used to ascertain the acid insoluble lignin. Then the amount of ash was determined by burning the solid residue at 575 °C in a muffle furnace. The acid soluble lignin was measured using a UV–Vis spectrophotometer at 240 nm with an extinction coefficient value of 106 $L \cdot g^{-1} \cdot cm^{-1}$. The lignin purity was calculated using the following equation:

$$\text{Lignin purity } (\%) = \frac{\text{Mass of acid soluble lignin} + \text{Mass of acid insoluble lignin}}{\text{Mass of the recovered lignin}} \times 100 \quad (2)$$

### 2.5. Measurement of solvatochromic parameters of DESs

A 10 mM stock solution of dye was prepared in methanol, and an appropriate amount of the dye stock solution was transferred to an Eppendorf tube. The methanol was removed via heating at 60 °C for 24 h. The DESs (1 mL) were transferred to the Eppendorf tube and sonicated at 25 °C to dissolve the dye. The absorbance of the solvents, which was less than 0.8, was analyzed via UV–Vis spectral scanning (probe **1**, *N*,*N*-diethyl-4-nitroaniline: 300–450 nm, probe **2**, Nile red: 400–650 nm, probe **3**, 4-nitroaniline: 300–450 nm) using an Optizen 3220 UV spectrophotometer (Mecasys, South Korea). The solvatochromic parameters were calculated using the following equations [1]:

$$E_{NR} = \frac{28591}{\lambda_{max}} \quad (3)$$

$$\pi^* = 0.314 \times (27.52 - \nu(1)) \quad (4)$$

$$\alpha = \frac{19.9657 - 1.0241\pi^* - \nu(2)}{1.6078} \quad (5)$$

$$\beta = 0.358 \times (31.10 - \nu(3)) - 1.125 \times \pi* \quad (6)$$

where $\nu(1)$, $\nu(2)$, and $\nu(3)$ represent the wavelength of the longest wavelength absorption band of the each probe in the region around 1000 $cm^{-1}$ and $\lambda_{max}$ [nm] represents the maximum wavelength due to the charge-transfer absorption band of Nile red (**2**).

### 2.6. Enzymatic hydrolysis of pretreated biomass

Enzymatic hydrolysis reactions were performed in 20 mL vials in a water bath at 120 rpm and 37 °C. Twenty milligrams of pretreated wood powder were subjected to cellulase hydrolysis for 24 h. The reaction was started by adding 16 μL Celluclast® to 1 mL of 50 mM sodium acetate buffer (pH 5.0) containing wood powder. After the saccharification period, the reaction mixtures were boiled for 3 min and centrifuged at 10,000 rpm for 20 min to remove any unhydrolyzed substrate. The amount of reducing sugar released via enzymatic hydrolysis was measured using the dinitrosalicylic acid (DNS) method [25]. The reducing sugar yield was calculated using the following equation:

$$\text{Reducing sugar yield } (\%) = \frac{\text{Mass of released reducing sugar}}{\text{Mass of glucose in the sample}} \times 100 \quad (7)$$

### 2.7. Characterization of pretreated biomass

The particle size and crystallinity index (CrI) of the pretreated wood powder were measured to determine the physical properties of the pretreated biomass. The particle size was measured using a particle-size analyzer (Mastersizer 2000, Malvern, UK).

The crystallinity index was determined via X-ray diffraction (XRD). The samples were scanned using a D8 ADVANCE Diffractometer (Bruker, Billerica, MA, USA) at 40 kV and 40 mA with 2-theta (θ) scanning from 2° to 40° at a 0.1° width. The following relationship was used to calculate the crystallinity index [26].

$$\text{CrI } (\%) = \frac{I_{200} - I_{AM}}{I_{200}} \times 100 \quad (8)$$

where $I_{200}$ is the diffraction intensity at the peak position of 21.7° for cellulose I and $I_{AM}$ is the intensity at suitable locations for the amorphous background.

The FT-IR spectra of the samples were determined by an average of 16 scans per spectrum in the range 4000–600 $cm^{-1}$ (resolution of 4 $cm^{-1}$ and scanning interval of 1 $cm^{-1}$) using an FT/IR-4100 type A spectrometer (Jasco International Co. Ltd., Tokyo, Japan).

### 2.8. Molecular weight of recovered lignin

The molecular weights of the lignin extracted with DES and cellulolytic enzyme lignin (CEL) were measured using an Agilent 1200S (Santa Clara, USA) with a multi-angle light scattering detector (miniDAWN TREOS). The column was operated at ambient temperatures and eluted at a flow rate of 1 mL/min. The Mn and Mw distributions were calibrated using polystyrene standards. Tetrahydrofuran (THF) was used as a mobile phase, and the lignin samples were acetylated in a pyridine and acetic anhydride mixture (1:1) prior to analysis.

CEL was prepared by following a previous procedure with slight modifications [27]. The wood powder was milled with a planetary ball mill (FRITSCH, Pulverisette 6, Germany) with a 500 mL sintered corundum bowl and 500 g sintered corundum balls (5 mm in diameter). The milling was performed at 500 rpm at room temperature for 24 h. The milled wood powder was then subjected to cellulase and hemicellulase hydrolysis (Novozyme 50199, Novozymes Korea, South Korea) for 24 h at 37 °C to remove the carbohydrates. The solid residue after enzymatic hydrolysis was washed with distilled water and then freeze-dried. The lignin was extracted from the freeze-dried sample using 95% dioxane for 24 h at ambient temperatures.

## 3. Results and discussion

### 3.1. Pretreatment of lignocellulosic biomass using DESs

Twenty-five types of ChCl-based DESs containing acid, hydroxyl, amide, and binary HBDs were prepared and used to pretreat pine wood powder in this study (Table 1). All the prepared DESs could be used to extract glucan, hemicellulose, and lignin from pine wood powder. Table 2 shows the recovered weights and the chemical compositions of the solid residue after pretreating the wood powder using DESs. The recovered weights (%) after pretreatment using DESs with acid HBDs ranged from 47% to 71%, much lower than weights obtained after pretreatment using DESs with hydroxyl and amide HBDs. The recovered weights after pretreatment with DESs containing the binary (acid and hydroxyl) HBD were compared and linearly decreased with increasing the ratio of acid HBD. For example, the recovered weights for $ChCl_1$:$LA_0$:$EG_2$ (no. 9), $ChCl_1$:$LA_{0.4}$:$EG_{1.6}$ (no. 21), $ChCl_1$:$LA_{1.2}$:$EG_{0.8}$ (no. 25), and $ChCl_1$:$LA_2$:$EG_0$ (no. 4) were 95, 84, 67, and 55%, respectively. The DES-mediated pretreatment process extracted the hemicellulose and lignin more efficiently than extracting the glucan from wood powder. Therefore, the hemicellulose and lignin content in the

**Table 2**
The chemical composition of the solid residue after DES pretreatment.

| DES | Recovered weight (%) | Glucan (%) | Hemicellulose (%) | Lignin (%) | Ash (%) |
|---|---|---|---|---|---|
| Untreated | 100 | 39.7 ± 0.6 | 22.1 ± 0.2 | 28.1 ± 0.1 | 1.0 ± 0.4 |
| ChCl:FA | 46.6 ± 2.3 | 69.3 ± 3.1 | 6.5 ± 2.6 | 18.0 ± 3.6 | 0.3 ± 0.2 |
| ChCl:AA | 66.5 ± 4.2 | 52.0 ± 3.4 | 15.5 ± 2.3 | 24.3 ± 0.4 | 0.6 ± 0.1 |
| ChCl:PA | 70.6 ± 1.7 | 49.6 ± 0.9 | 14.8 ± 1.7 | 23.3 ± 0.2 | 0.3 ± 0.2 |
| ChCl:LA | 54.6 ± 5.4 | 60.4 ± 6.4 | 10.6 ± 1.4 | 16.5 ± 0.3 | 0.3 ± 0.1 |
| ChCl:CA | 61.4 ± 0.6 | 50.2 ± 1.1 | 8.5 ± 1.6 | 27.1 ± 0.9 | 0.5 ± 0.3 |
| ChCl:MA | 56.7 ± 1.5 | 58.1 ± 1.2 | 6.4 ± 0.2 | 25.8 ± 1.2 | 0.3 ± 0.0 |
| ChCl:SA | 66.6 ± 4.4 | 51.2 ± 2.5 | 10.7 ± 0.2 | 24.3 ± 2.4 | 0.4 ± 0.4 |
| ChCl:G | 94.6 ± 1.1 | 40.1 ± 2.3 | 19.9 ± 1.2 | 25.5 ± 1.0 | 0.6 ± 0.2 |
| ChCl:EG | 95.2 ± 4.4 | 41.0 ± 1.6 | 22.6 ± 0.7 | 25.2 ± 1.5 | 0.4 ± 0.4 |
| ChCl:DG | 92.0 ± 1.6 | 41.1 ± 3.0 | 20.2 ± 0.5 | 24.8 ± 0.3 | 0.5 ± 0.3 |
| ChCl:TG | 97.8 ± 2.2 | 40.1 ± 1.5 | 22.4 ± 0.7 | 25.3 ± 0.8 | 0.5 ± 0.3 |
| ChCl:X | 93.1 ± 2.6 | 39.9 ± 2.7 | 19.8 ± 0.8 | 25.2 ± 1.1 | 0.5 ± 0.3 |
| ChCl:U | 99.7 ± 0.1 | 37.0 ± 0.3 | 17.5 ± 1.4 | 23.2 ± 0.3 | 0.5 ± 0.3 |
| ChCl:F | 92.5 ± 2.3 | 41.1 ± 0.4 | 20.8 ± 0.4 | 23.7 ± 0.0 | 0.7 ± 0.0 |
| ChCl:A | 94.1 ± 3.0 | 41.5 ± 2.5 | 23.0 ± 1.3 | 25.8 ± 0.5 | 0.5 ± 0.2 |
| ChCl:LA:FA | 52.5 ± 2.6 | 63.4 ± 4.1 | 10.3 ± 2.1 | 15.2 ± 0.6 | 0.5 ± 0.3 |
| ChCl:LA:F | 86.5 ± 3.6 | 41.4 ± 0.2 | 17.6 ± 1.0 | 24.5 ± 0.3 | 0.7 ± 0.0 |
| $ChCl_1:FA_{0.4}:EG_{1.6}$ | 92.9 ± 0.2 | 50.8 ± 1.5 | 15.5 ± 1.0 | 21.6 ± 1.5 | 0.7 ± 0.0 |
| $ChCl_1:AA_{0.4}:EG_{1.6}$ | 93.3 ± 3.2 | 40.0 ± 1.1 | 23.7 ± 1.4 | 25.1 ± 1.6 | 0.5 ± 0.3 |
| $ChCl_1:PA_{0.4}:EG_{1.6}$ | 88.8 ± 1.5 | 41.4 ± 0.4 | 23.9 ± 1.6 | 23.9 ± 0.5 | 0.7 ± 0.0 |
| $ChCl_1:LA_{0.4}:EG_{1.6}$ | 83.8 ± 1.3 | 44.0 ± 1.0 | 21.7 ± 1.2 | 24.0 ± 0.5 | 0.6 ± 0.2 |
| $ChCl_1:FA_{1.2}:EG_{0.8}$ | 63.7 ± 0.0 | 53.1 ± 0.7 | 14.0 ± 0.1 | 21.3 ± 1.9 | 0.5 ± 0.2 |
| $ChCl_1:AA_{1.2}:EG_{0.8}$ | 81.6 ± 2.2 | 41.0 ± 1.8 | 18.5 ± 1.0 | 24.5 ± 0.9 | 0.5 ± 0.3 |
| $ChCl_1:PA_{1.2}:EG_{0.8}$ | 86.0 ± 0.6 | 41.4 ± 1.2 | 20.5 ± 1.1 | 25.1 ± 1.5 | 0.5 ± 0.2 |
| $ChCl_1:LA_{1.2}:EG_{0.8}$ | 67.1 ± 3.8 | 50.4 ± 3.0 | 14.6 ± 1.0 | 21.8 ± 0.2 | 0.7 ± 0.0 |

solid residue after DES pretreatment were lower than those in the untreated wood powder, while the glucan contents in the solid residue were higher than those in the untreated wood powder. The ash content in the solid residue after DES pretreatment was about 30–70% of that in the untreated wood powder. Although the content of ash in the wood powder was quite small, the ash was also removed by the DES pretreatment process.

Fig. 1 shows the removal ratios of glucan, hemicellulose, and lignin from the wood powder pretreated with DESs. The DESs containing acid HBDs effectively extracted hemicellulose and lignin from the wood powder, with the removal ratios of hemicellulose ranging from 53% to 86%, much higher than the removal ratios obtained by DESs containing hydroxyl and amide HBDs (Fig. 1b). Among all the DESs tested, ChCl:FA was the most efficient DES to extract hemicellulose from the wood powder. The hemicellulose removal by DESs containing binary HBDs increased with the increase in the ratio of acid HBD. For example, hemicellulose removal for $ChCl_1:PA_0:EG_2$ (no. 9), $ChCl_1:PA_{0.4}:EG_{1.6}$ (no. 20), $ChCl_1:PA_{1.2}:EG_{0.8}$ (no. 24), and $ChCl_1:PA_2:EG_0$ (no. 3) were 8, 9, 20, and 53%, respectively. The removal ratios of lignin from the wood powder by using DESs containing acid HBDs ranged from 41% to 70%, which is much higher than those obtained by DESs containing hydroxyl and amide HBDs (Fig. 1c). The ChCl:FA and ChCl:LA could efficiently extract lignin from the wood powder. For the DESs containing binary HBDs, the lignin removal increased with the increase in the ratio of acid HBD. For example, the lignin removal for $ChCl_1:FA_0:EG_2$ (no. 9), $ChCl_1:FA_{0.4}:EG_{1.6}$ (no. 18), $ChCl_1:FA_{1.2}:EG_{0.8}$ (no. 22), and $ChCl_1:FA_2:EG_0$ (no. 1) were 14, 19, 52, and 70%, respectively. ChCl:LA:FA was the most efficient DES in extracting lignin, with 72% removal. Interestingly, the use of the binary HBDs LA:FA (1:1) could synergistically enhance the extraction capacity for lignin compared with ChCl:FA and ChCl:LA.

The removal ratio of hemicellulose correlated well with the that of lignin with a high correlation coefficient ($R = 0.919$, $n = 25$). This may indicate that the links between the hemicellulose and lignin were cleaved by the DES, and each compound had become dissolved in the DES. DESs are known to have a strong internal hydrogen bonding network, and, because of this property, the bonds between hemicellulose and lignin can be broken [1]. The removal ratios of glucan from the wood powder using DESs containing acid HBDs ranged from 12% to 23%, which is higher than the removal ratios obtained by DESs containing hydroxyl and amide HBDs (Fig. 1a). However, the removal ratio of glucan after DES pretreatment was much lower than those of hemicellulose and lignin. It is widely known that the solubility of cellulose in DES is extremely low [12,28]. Considering the results of previous studies, it might be that glucose was mostly eluted from the hemicelluloses dissolved in the DES rather than cellulose. The high correlation coefficient ($R = 0.900$, $n = 25$) between the hemicellulose removal and glucan removal agrees with this conclusion. As a result of the DES-mediated pretreatment of pine wood powder, the glucan content in the pretreated solid residue increased regardless of the type of DES used. This result is due to the removal of significant amounts of hemicellulose and lignin in the pretreatment process using the DES.

### 3.2. Influence of DES characteristics on the pretreatment of lignocellulosic biomass

Understanding the solvent properties of DESs is helpful to explain their effects in the pretreatment of lignocellulosic biomass. The solvatochromic parameters of the ChCl-based DESs with various HBDs were evaluated in this study (Table 3). The $E_{NR}$ value quantifies the polarity of a solvent using Nile red dye. The Kamlet–Taft solvatochromic parameters are the most comprehensive and commonly used quantitative measurements of solvent properties [18,29]. Three solvatochromic parameters exist: $\pi^*$, $\alpha$, and $\beta$, which measure polarity/polarizability, hydrogen bond donating ability (acidity), and hydrogen bond accepting ability (basicity), respectively.

The $E_{NR}$ values of the DESs in this study ranged from 47.6 to 50.9 and were lower than those of short-chain alcohols (the value for methanol is 52.0). These results indicate that the polarity of the DESs studied is higher than that of methanol and similar to that of water ($E_{NR} = 48.2$). Among the studied DESs, ChCl:CA exhibited the highest polarity, whereas ChCl:F exhibited the lowest. The polarity of the ChCl-based DESs increased according to the HBD partner in the following order:

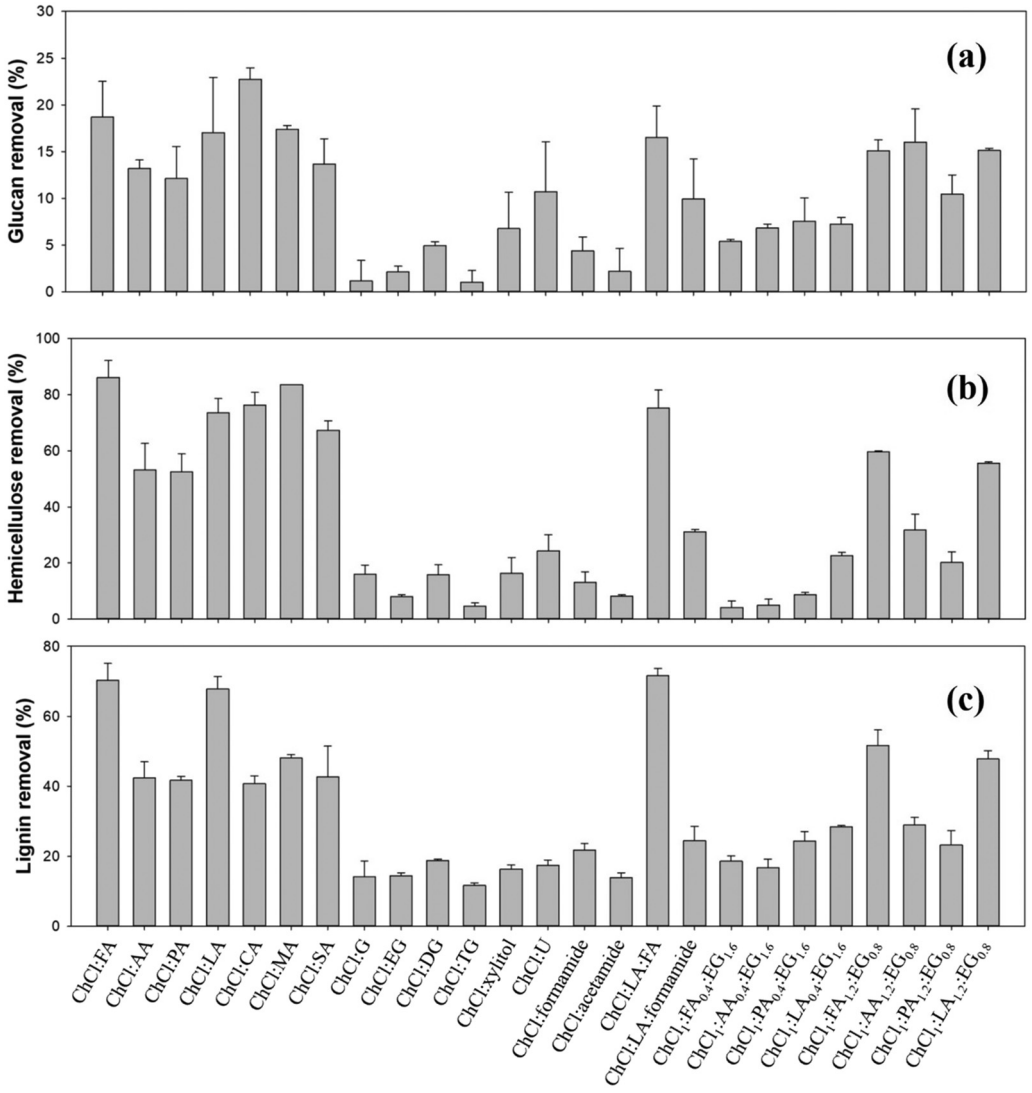



**Fig. 1.** Component loss of pine wood powder after DES pretreatment. (a): glucan removal, (b): hemicellulose removal, (c): lignin removal.

F < DG < EG < TG < U < G ≈ X < PA < AA < LA < SA < MA < FA < CA. The polarity of the DESs containing acid HBD was higher than those containing amide or hydroxyl HBD used in this study as well as those previously reported for general ILs and organic solvents.

The solvatochromic parameter $\pi^*$ represents the combination of the dipolarity and polarizability of the solvents. The $\pi^*$ values of the studied DESs ranged from 0.968 to 1.213 and were much higher than that of methanol ($\pi^* = 0.60$). Most DESs exhibit similar $\pi^*$ values to dimethyl sulfoxide ($\pi^* = 1.00$) and water ($\pi^* = 1.09$). Among the DESs tested, ChCl:U exhibited the highest $\pi^*$ value, whereas ChCl:PA exhibited the lowest. The $\pi^*$ values of the ChCl-based DESs increased according to the HBD partner in the following order: PA < AA < TG < DG < MA < EG < FA < LA ≈ SA < CA < F < G < X < U.

The $\alpha$ value represents the hydrogen bond acidity normalized to a value of unity for methanol. The $\alpha$ values of the DESs examined covered a wide range from 0.606 to 1.344, ranging from that of *tert*-butanol ($\alpha = 0.68$) through water ($\alpha = 1.17$). Among the studied DESs, ChCl:LA:FA exhibited the highest $\alpha$ value, whereas ChCl:F exhibited the lowest $\alpha$ value. For the different HBD partners, the $\alpha$ values of the DESs increased in the following order: F < DG < EG < U < TG < X < G < PA < LA < SA < AA < MA < FA < CA.

The $\beta$ value represents the solvent hydrogen bond basicity normalized to a value of unity for hexamethylphosphoramide. The $\beta$ values of the DESs studied in this work ranged from 0.387 to 0.684, or from the values for dioxane ($\beta = 0.38$) through *N*,*N*-dimethylformamide ($\beta = 0.69$). Among the studied DESs, ChCl:DG exhibited the highest $\beta$ value, whereas ChCl:FA exhibited the lowest. The DESs with a glycol HBD exhibited high $\beta$ values, whereas the DESs with acid HBD exhibited poor ability to accept hydrogen bonds [18]. The $\beta$ values of the DESs increased according to the HBD partner in the following order: FA < CA < LA < SA ≈ G < U < AA < PA < F < X < EG < MA < TG < DG.

We also analyzed the relationship between the solvatochromic parameters of the DESs and their pretreatment efficiency for pine wood powder. When the four solvent parameters of all the studied DESs were correlated with the recovered weight of the solid residue after pretreatment, good correlations were obtained for the $E_{NR}$ value ($R = 0.913$, $n = 24$) and hydrogen bond acidity ($R = 0.924$, $n = 24$). The recovered weight increased with an increase in the $E_{NR}$ value, indicating that the high polarity of DESs is beneficial for extracting wood components from pine wood powder. The recovered weight of solid residue decreased with the increase in $\alpha$ value of DESs, which means that the DESs with high acidity can efficiently extract wood components from pine wood powder.

When the solvatochromic parameters of the DESs were correlated with the removal ratios of glucan, hemicellulose, and lignin from the pine wood powder after pretreatment with DESs, the $E_{NR}$ and $\alpha$ values

**Table 3**
Solvatochromic parameters of prepared DESs.

| DES | $E_{NR}$ | $\pi^*$ | $\alpha$ | $\beta$ |
|---|---|---|---|---|
| ChCl:FA | 47.9 | 1.111 | 1.293 | 0.387[a] |
| ChCl:AA | 48.5 | 0.985 | 1.232 | 0.543 |
| ChCl:PA | 48.8 | 0.968 | 1.193 | 0.556 |
| ChCl:LA | 48.3 | 1.121 | 1.196 | 0.477 |
| ChCl:CA | 47.6 | 1.123 | 1.339 | 0.417[a] |
| ChCl:MA | 48.0 | 1.095 | 1.277 | 0.654 |
| ChCl:SA | 48.2 | 1.121 | 1.226 | 0.518[a] |
| ChCl:G | 50.0 | 1.155 | 0.797 | 0.518 |
| ChCl:EG | 50.7 | 1.109 | 0.680 | 0.614 |
| ChCl:DG | 50.8 | 1.072 | 0.676 | 0.684 |
| ChCl:TG | 50.5 | 1.055 | 0.765 | 0.678 |
| ChCl:X | 50.0 | 1.205 | 0.769 | 0.589 |
| ChCl:U | 50.3 | 1.213 | 0.710 | 0.519 |
| ChCl:F | 50.9 | 1.144 | 0.606 | 0.588 |
| ChCl:A | ND[b] | ND | ND | ND |
| ChCl:LA:FA | 47.7 | 1.111 | 1.344 | 0.487[a] |
| ChCl:LA:F | 50.6 | 1.136 | 0.695 | 0.549 |
| $ChCl_1$:$FA_{0.4}$:$EG_{1.6}$ | 50.3 | 1.109 | 0.761 | 0.560 |
| $ChCl_1$:$AA_{0.4}$:$EG_{1.6}$ | 50.7 | 1.094 | 0.700 | 0.617 |
| $ChCl_1$:$PA_{0.4}$:$EG_{1.6}$ | 50.7 | 1.099 | 0.691 | 0.603 |
| $ChCl_1$:$LA_{0.4}$:$EG_{1.6}$ | 49.3 | 1.112 | 0.986 | 0.515 |
| $ChCl_1$:$FA_{1.2}$:$EG_{0.8}$ | 48.9 | 1.116 | 1.062 | 0.520 |
| $ChCl_1$:$AA_{1.2}$:$EG_{0.8}$ | 50.8 | 1.020 | 0.724 | 0.550 |
| $ChCl_1$:$PA_{1.2}$:$EG_{0.8}$ | 50.6 | 1.086 | 0.720 | 0.592 |
| $ChCl_1$:$LA_{1.2}$:$EG_{0.8}$ | 48.6 | 1.101 | 1.140 | 0.590 |

[a] Estimated by extrapolation.
[b] Not determined.

of DESs greatly influenced the removal efficiency of wood components. The best correlation was obtained for the hemicellulose removal. The removal ratio of hemicellulose increased with decreasing the $E_{NR}$ value ($R = 0.928$, $n = 24$) and increasing the $\alpha$ value ($R = 0.925$, $n = 24$) of DES (Fig. 2a). The removal ratio of lignin also increased with decreasing the $E_{NR}$ value ($R = 0.858$, $n = 24$) and increasing the $\alpha$ value ($R = 0.863$, $n = 24$) of DES (Fig. 2b).

The removal ratio of hemicellulose and the removal ratio of lignin correlated well ($R = 0.919$, $n = 25$), and they were highly influenced by the same parameters ($E_{NR}$ and $\alpha$). The removal ratio of glucan was well correlated with the $E_{NR}$ ($R = 0.777$, $n = 24$) and $\alpha$ values ($R = 0.781$, $n = 24$) among the four parameters, but the correlation coefficient for glucan removal was much lower than those for hemicellulose and lignin removal. Among the four parameters that indicate the DES properties, the hydrogen bond acidity ($\alpha$) was found to have the greatest influence on the pretreatment efficiency for pine wood powder. The principle for the extraction of biomass components is the strong hydrogen bonding between the DESs and biomass [1,30]. Alvarez-Vasco and coworkers suggested that DES may provide a mild acid-base catalysis mechanism which can catalyze the controlled cleavage of labile ether linkages among phenylpropane units of lignin [13]. Shen et al. reported that ChCl:LA (1:10) has selectively cleaved the $\beta$-*O*-4 ether bonds without affecting the C—C bonds in lignin of *Eucalyptus* by the analysis of 2D HQSC NMR [31]. Recently, da Costa lopes and coworkers studied the cleavage of the $\beta$-*O*-4 ether bond present in 2-phenoxy-1-phenylethanol (PPE) as a lignin model compound by using PA:U (2:1), ChCl:LA (1:10), and ChCl:*p*-toluene sulfonic acid (1:1) [32]. The high acidity of ChCl:*p*-toluene sulfonic acid (1:1) induced the cleavage of PPE, although the DES caused the undesired condensation of cleavage product. With similar mechanism, our results could be understood. DES which has higher hydrogen bond acidity may cause the higher extraction efficiency of lignin by catalyzing the cleavage of $\beta$-*O*-4 ether bond.

### 3.3. Characterization and enzymatic hydrolysis of pretreated biomass

Pine wood powder pretreated with DESs was enzymatically hydrolyzed by cellulase. An increase in the enzymatic saccharification of biomass leads to an increase in biomass utilization through microbial fermentation. Thus, it is important to measure the enzymatic saccharification of biomass to estimate pretreatment efficiency. In this study, the biomass pretreated with the DES, except ChCl:CA, exhibited higher reducing sugar yields, compared with untreated biomass (Fig. 3).

For the different HBD partners, the reducing sugar yield of the biomass pretreated with DESs increased in the following order: CA < hydroxyl HBDs ≈ MA ≈ PA < amide HBDs ≈ AA < SA < FA < LA. The biomass pretreated with DESs containing hydroxyl HBD and amide HBD showed about 1.2 and 1.3 times higher reducing sugar yield, respectively, compared with untreated biomass. Interestingly, after pretreatment with ChCl:LA:FA, ChCl:LA, and ChCl:FA, the reducing sugar

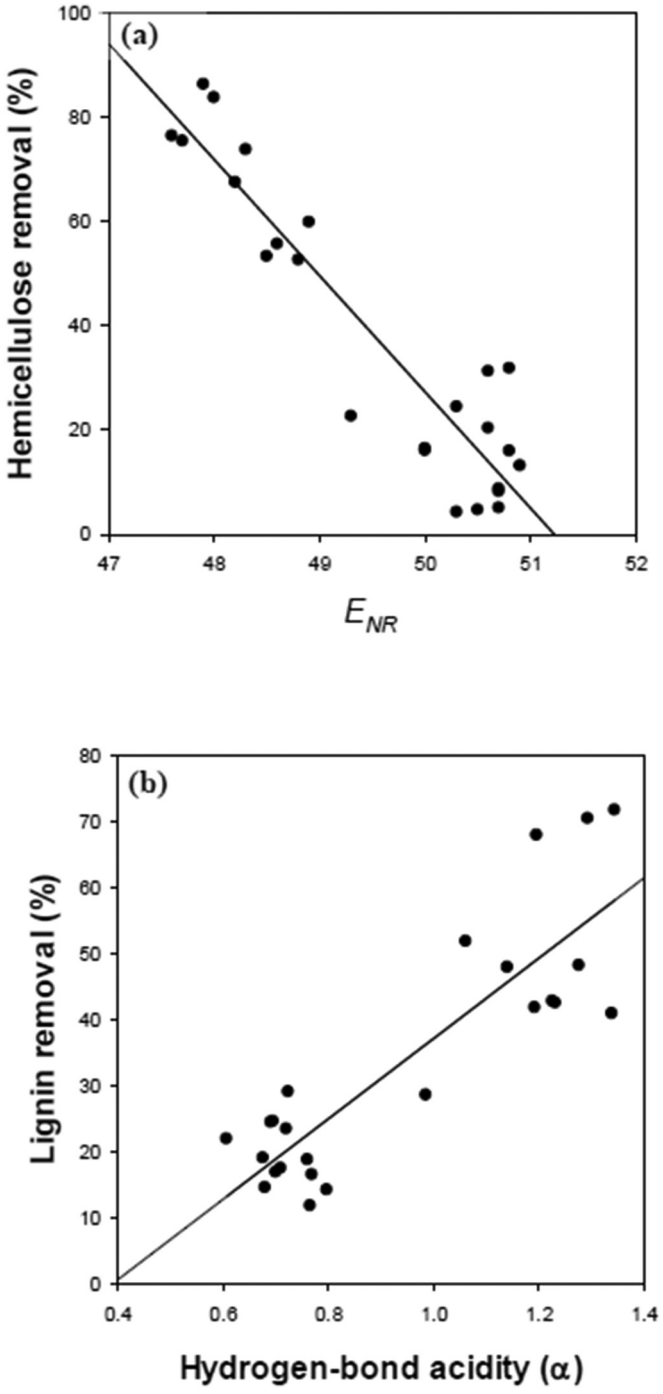



**Fig. 2.** Correlation between the solvatochromic parameter of DESs and removal efficiency of hemicellulose (a) and lignin (b).

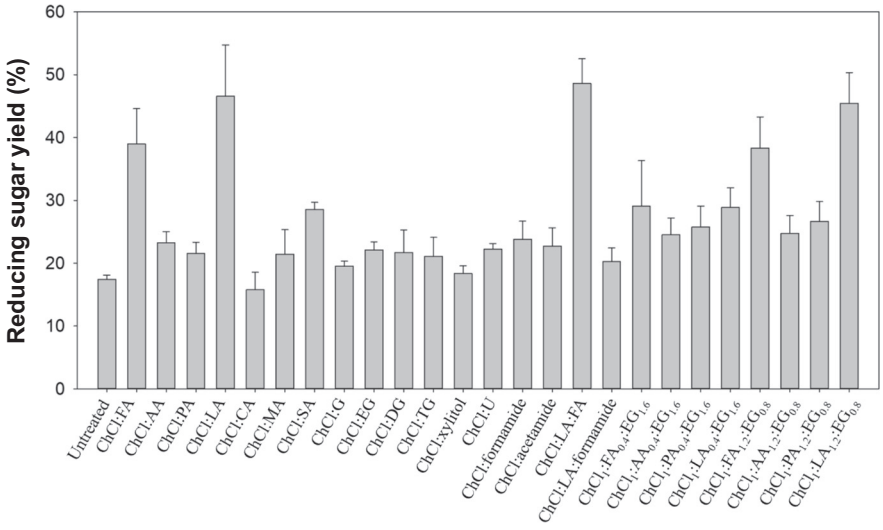



**Fig. 3.** Enzymatic hydrolysis of untreated and DES-treated wood powder.

yield of the pretreated biomass was significantly increased to 2.7, 2.6, and 2.2 times higher than that of the untreated biomass, respectively. Biomass pretreated with ChCl:LA:FA exhibited higher enzymatic saccharification than that using DES containing a single HBD, which suggests that the use of binary HBDs offers advantages. When the reducing sugar yield of biomass pretreated with DESs containing binary HBDs of

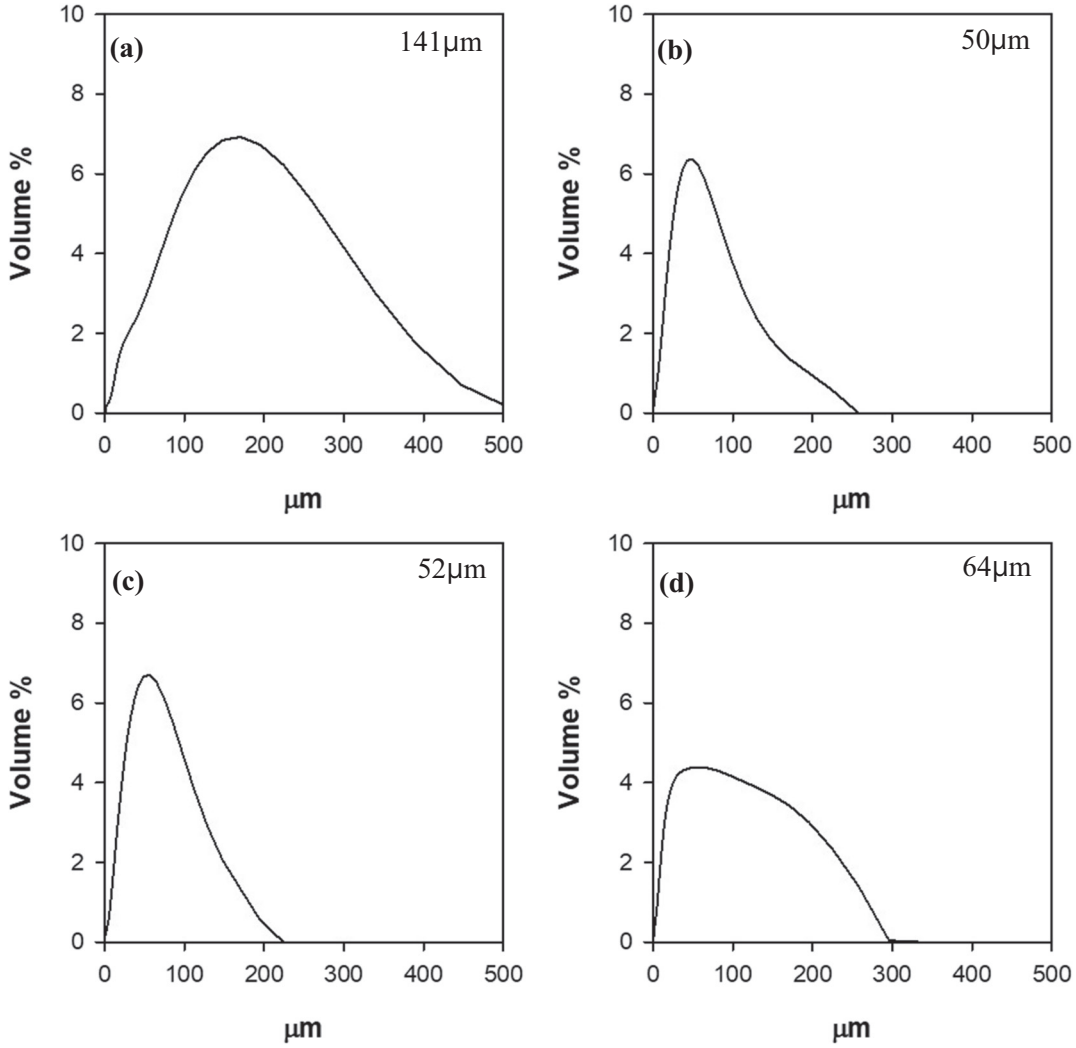
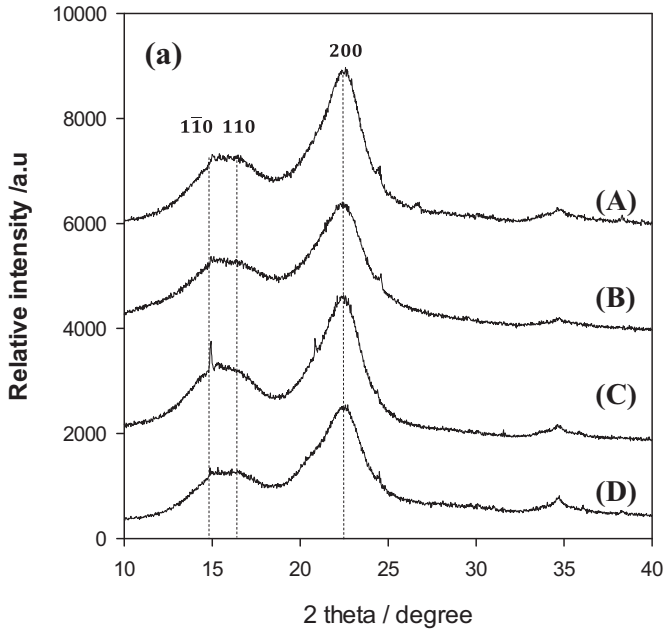



**Fig. 4.** Particle size distribution of untreated and wood powder pretreated with DESs. (a): untreated, (b): ChCl:LA treated, (c): ChCl:FA treated, (d): ChCl:LA:FA treated.



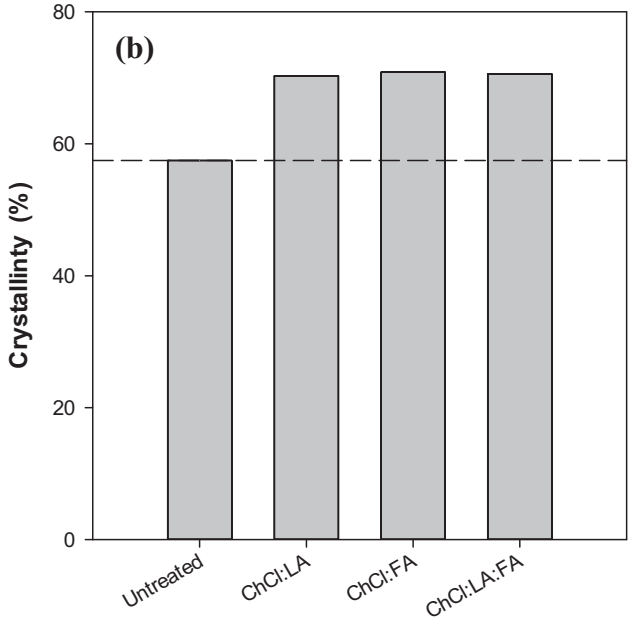
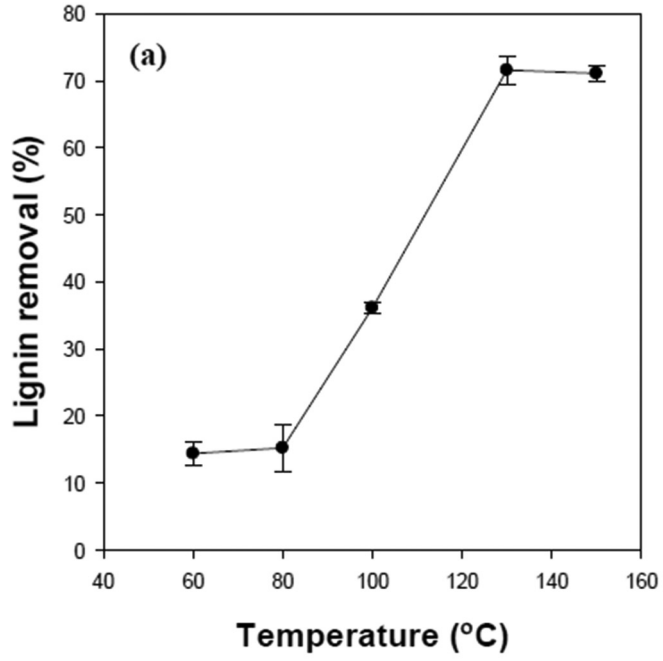




**Fig. 5.** XRD pattern (a) and crystallinity index (b) of untreated and wood powder pretreated with DESs. (A): untreated, (B): ChCl:LA treated, (C): ChCl:FA treated, (D): ChCl:LA:FA treated.



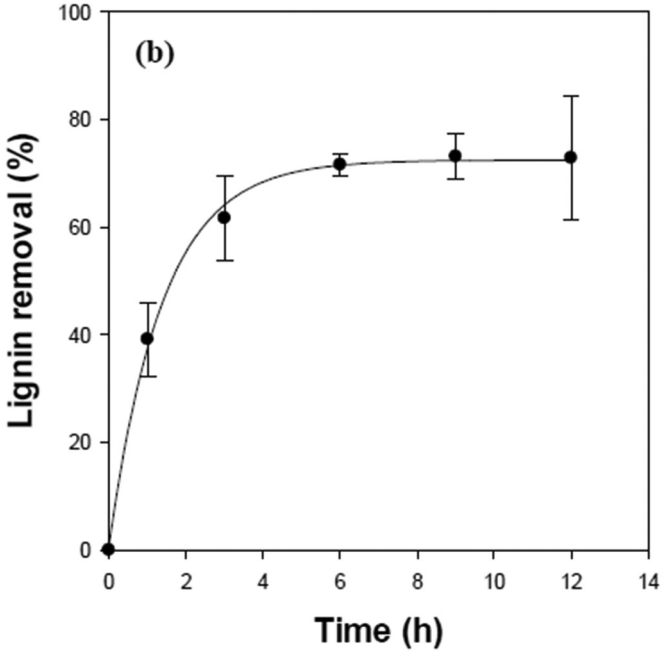




**Fig. 6.** Effect of pretreatment temperature (a) and incubation time (b) on the lignin extraction from wood powder using ChCl:LA:FA.

acid and hydroxyl HBD were compared, the reducing sugar yield increased with increasing the ratio of acid HBD. For example, the reducing sugar yield of biomass pretreated with $ChCl_1$:$LA_0$:$EG_2$ (no. 9), $ChCl_1$:$LA_{0.4}$:$EG_{1.6}$ (no. 21), $ChCl_1$:$LA_{1.2}$:$EG_{0.8}$ (no. 25), and $ChCl_1$:$LA_2$:$EG_0$ (no. 4) were 1.3, 1.7, 2.6, and 2.7 times higher than that of untreated biomass, respectively. Conversely, pretreatment with DESs containing binary HBDs of acid (LA) and amide HBD (F) resulted in lower reducing sugar yields than the pretreatment with DESs containing a single HBD.

It is widely known that the lignin in biomass inhibits the action of cellulase. Likewise, we found that the higher the lignin removal ratio in wood powder, the better the enzymatic saccharification of biomass. There was a high correlation between lignin removal and the saccharification yield, with a high correlation coefficient value ($R$) of 0.751 ($n = 25$). Additionally, the saccharification of cellulose is affected by various factors, such as the chemical composition of the biomass, crystallinity and degree of the polymerization of cellulose, and specific surface area [33]. In this study, we analyzed the physical properties of the pretreated wood powder to determine the cause for the observed increase in enzymatic saccharification. This study was conducted using wood powder pretreated with DESs that exhibited high pretreatment efficiency (ChCl:LA, ChCl:FA, and ChCl:LA:FA). For all three pretreated samples, particle size reduction was evident (Fig. 4). The average particle sizes of biomasses pretreated with ChCl:LA, ChCl:FA, and ChCl:LA:FA were 50, 52, and 64 μm, respectively, which were much smaller than that of untreated biomass (141 μm). This could be due to the removal of

hemicellulose and lignin from the biomass in the pretreatment process. The pretreatment caused a reduction in the particle size of cellulose and therefore increased the externally available surface area. Increasing the surface area of the biomass could enhance the affinity between cellulose and enzymes and thus increase the hydrolysis rate [34,35]. The structural changes after pretreatment with DESs were assessed via XRD experiments. Fig. 5 shows the XRD patterns and crystallinity of biomass pretreated with DESs. The biomass exhibited the three expected peaks at $2\theta = 15.2°$, 16.4°, and 22.6°, which correspond to the $(1\bar{1}0)$, (110), and (200) crystalline planes of cellulose I [36]. Notably, the CrI of each type of biomass crystallinity increased after pretreatment. A reduction in the crystallinity of cellulose was reported to be a possible cause behind the increased enzymatic saccharification, but the increases in crystallinity were observed for pretreated biomass. We believe that the increases in the CrI values can be ascribed primarily to the removal of amorphous hemicellulose and lignin, leading to an increase in the crystalline substrate in the biomass. This is evident by the data reported in Table 2, which showed an overall higher glucan content after pretreatment [37].

### 3.4. Influence of pretreatment temperature and time on biomass

Considering the lignin removal ratio and enzymatic saccharification yield after pretreatment, ChCl:LA:FA was found to be the most effective DES for biomass pretreatment among the DESs studied. Therefore, ChCl:LA:FA was selected as the pretreatment solvent to identify the optimum pretreatment conditions. To find the optimum temperature for pretreatment, wood powder was pretreated with the DES for 6 h at different temperatures ranging from 60 °C to 150 °C (Fig. 6a). When the temperature was increased from 60 °C to 80 °C, the lignin removal ratio did not increase significantly. However, the lignin removal increased sharply as the pretreatment temperature was increased from 80 °C to 100 °C and 130 °C. However, when the pretreatment temperature was increased to 150 °C, no differences in the lignin removal ratio were observed compared with the biomass pretreated at 130 °C. The pretreatment time was varied between 1 and 12 h for analysis. In these experiments, the pretreatment temperature was fixed at 130 °C (Fig. 6b). The lignin removal ratio increased with increasing the pretreatment time up to 6 h. However, after 6 h, the lignin removal ratio reached equilibrium. Overall, these results suggest that pretreatment at 130 °C for 6 h is optimal.

### 3.5. Recovery of extracted lignin

After the DES-mediated pretreatment process to extract lignin, ethanol was added to the treated wood to separate the solid residues from the DES. The DES–ethanol solution was separated via filtration, and the ethanol was removed using a rotary evaporator. The soluble fraction mostly contains hemicellulose, lignin, and DES. Lignin can be easily obtained from soluble fraction by precipitation using water as an anti-solvent at low temperature. The principle for the separation of hemicellulose and lignin is that the solubility of lignin decreases as the water content in the DES increases. The recovery yield and purity of lignin were measured when the pine wood powder was pretreated with acid DESs, which exhibited high lignin extraction efficiency (Fig. 7). The DESs that exhibited high lignin extraction efficiency were also effective for recovering lignin from the DESs. The yields of lignin extracted with ChCl:FA, ChCl:LA, and ChCl:LA:FA were 61%, 53%, and 54%, respectively. Because the ChCl ions in the DESs bind to the phenol group of lignin, it is necessary to use a solvent that can effectively separate the lignin and DESs to increase the purity of the recovered lignin [38]. In this study, the lignin was washed with a water/ethanol (9:1, v/v) mixture, which has been reported to be effective for lignin and DES separation [13]. The purity of the recovered lignin was not significantly affected by the type of HBD used and reached high values of 90–97%.

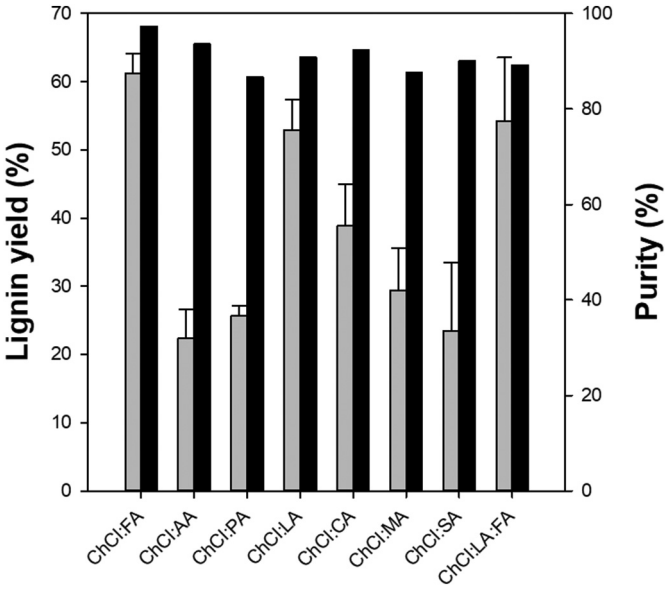



**Fig. 7.** The lignin yield (gray bar) and the purity (black bar) of precipitated lignin.

The FT-IR spectra of lignin prepared using ChCl:LA:FA was investigated by comparing with wood powder and pretreated wood powder (Fig. 8). In the fingerprint region, well known many absorption peaks appeared [39–41]. The positions of most characteristic bands for carbohydrates are 898, 1034, 1055, 1105, 1160, and 1315 $cm^{-1}$. Those absorption peaks were more pronounced in the pretreated wood powder. The band around 898 $cm^{-1}$ is the fingerprint region representing the $\beta$-(1,4)-glycosidic bond in cellulose. The band 1034, 1055 and 1105 $cm^{-1}$ are observed from C—O stretching in carbohydrate [39]. Then, the area of the bands 1160 and 1315 $cm^{-1}$ arises mainly from C—O—C asymmetric stretching in cellulose and $CH_2$ wagging in crystalline cellulose, respectively [40]. On the other hand, the FTIR spectra of extracted lignin with ChCl:LA:FA showed the

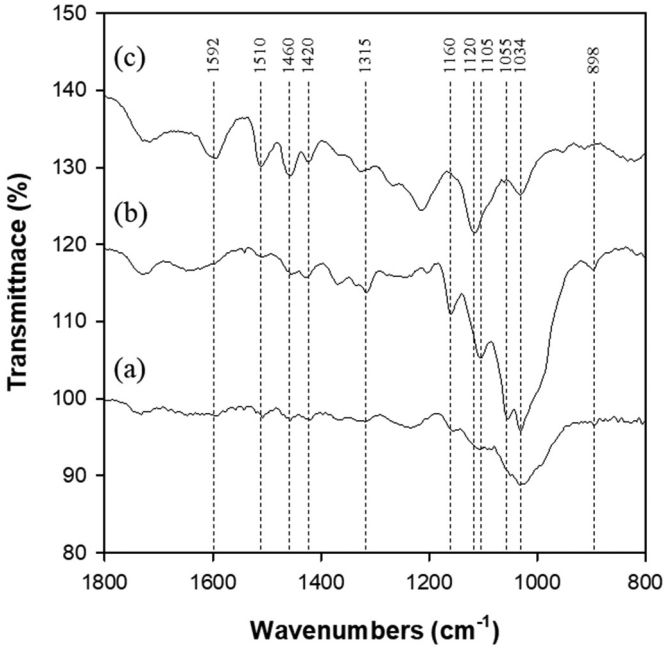



**Fig. 8.** FT-IR spectra of untreated wood powder (a), pretreated wood powder (b), and lignin extracted with ChCl:LA:FA (c).

**Table 4**
Molecular weights and polydispersity of lignins.

| | Lignin extracted with ChCl:LA:FA (1:1:1) | Cellulolytic enzyme lignin |
|---|---|---|
| $M_w$ | 1801 | 12,756 |
| $M_n$ | 1218 | 7211 |
| PDI ($M_w/M_n$) | 1.48 | 1.77 |

characteristics of lignin. The main absorption bands of extracted lignin with ChCl:LA:FA include 1120, 1420, 1510, and 1592 $cm^{-1}$ (aromatic skeletal vibrations), and 1460 $cm^{-1}$ (asymmetric C—H deformations in lignin) [41].

The molecular weights of lignin prepared using ChCl:LA:FA and cellulolytic enzyme lignin (CEL) as a standard of native lignin were determined via gel-permeation chromatography (GPC). The weight-average (Mw) and number-average (Mn) molecular weights are listed in Table 4. The calculated polydispersity indices of DES–lignin (ChCl:LA:FA) and CEL were 1.48 and 1.77, respectively. Compared with CEL, the DES–lignin (ChCl:LA:FA) had a significantly lower molecular weight.

### 3.6. Recycling of DES used for biomass pretreatment

The advantageous characteristics of DES distinguished from conventional solvents are low toxicity, nonvolatility, low cost, and recyclability. To develop the environmental-friendly biorefinery process using DES, the recyclability of DES should be considered. The ChCl:LA:FA, the most efficient DES to extract lignin from wood powder, was repeatedly used to investigate the efficiency of DES recovery and retainment of lignin extraction capacity (Fig. 9). The ChCl:LA:FA used for biomass pretreatment could be recovered by evaporation of water due to the very low volatility and thermal stability of ChCl:LA:FA under 100 °C. The water content (3.96%) of recovered ChCl:LA:FA was lower than the initial water content (6.12%) in prepared ChCl:LA:FA. The extremely low volatility of ChCl:LA:FA is very interesting, because the boiling points of lactic acid and formic acid are about 100 °C. It could be understood by the strong hydrogen bonding networks to form DES between HBA (ChCl) and HBD (LA and FA) [42]. Over 90% of ChCl:LA:FA could be recovered after 5-time reuse. Over 82% of delignification capacity of ChCl:LA:FA could be also retained after 5-time reuse. The first lignin removal yield by fresh ChCl:LA:FA was 72% and the lignin removal yield by 5-time reused ChCl:LA:FA was 59%. Decreased delignification capacity of reused ChCl:LA:FA may be caused by the accumulation of hemicellulose. The concentrated hemicelluloses after repeated use of ChCl:LA:FA could be removed by precipitation using anti-solvents or by membrane dialysis. The lignin extraction yield could be enhanced after the purification of ChCl:LA:FA by removing hemicelluloses.

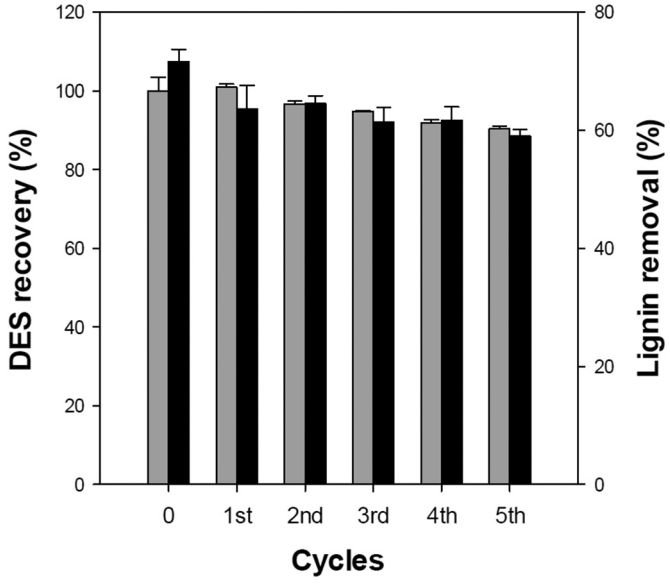



**Fig. 9.** Recycling of ChCl:LA:FA used for the extraction of lignin. Gray bar and black bar represent the recovery of ChCl:LA:FA and efficiency of lignin removal after reuse, respectively.

## 4. Conclusion

Various ChCl-based DESs containing acid, hydroxyl, amide, and binary HBDs were prepared and successfully used to pretreat pine wood powder in this work. Among the studied DESs, those with acid HBDs were the most efficient to extract hemicellulose and lignin. As a result of the DES-mediated pretreatment of biomass, the resulting content of glucan increased, whereas the contents of hemicellulose and lignin were efficiently decreased. The correlation between the solvent properties of DESs and the pretreatment efficiency for biomass demonstrated that the polarity and hydrogen bond acidity of the DESs are the most important factor to extract hemicellulose and lignin from biomass. These results are meaningful to understand the extraction mechanism of hemicellulose and lignin from biomass by using DESs. In particular, ChCl:LA:FA, with the highest $\alpha$ value among the tested DESs, most efficiently extracted lignin from biomass. After pretreatment with DESs, the enzymatic saccharification yield of biomass was greatly enhanced. The increased reducing sugar yield could be explained by the decreased particle size and the reduced contents of hemicellulose and lignin in the biomass. During the pretreatment process of biomass using DESs, the extracted lignin could be recovered successfully with high yield and high purity. In addition, the DES used for pretreatment process could be also successfully reused with high recovery yield of DES and high retention of delignification capacity. Therefore, the results of this study will open up new opportunities for various applications of DESs for biomass pretreatment and lignin extraction.

## CRediT authorship contribution statement

**Yujin Oh:** Conceptualization, Methodology, Visualization, Writing - Original Draft. **Saerom Park:** Conceptualization, Software, Formal analysis, Investigation, Data Curation. **Dahun Jung:** Validation, Resources, Visualization. **Kyeong Keun Oh:** Supervision, Project administration, Funding acquisition. **Sang Hyun Lee:** Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

## Acknowledgments

This work was supported by the C1 Gas Refinery Program [2020M3D3A1A01028904], the Technology Development Program to Solve Climate Changes [2017M1A2A2087647], and the Basic Science Research Program [2018R1D1A1B07050163] through the National Research Foundation of Korea (NRF) grant funded by the Ministry of Science and ICT. This study was also supported by the Korea Ministry of Environment as "The SEM (Subsurface Environment Management) projects" [2020002480009]. Partly, this paper was supported by Konkuk University Researcher Fund in 2020.

## References

[1] Y. Liu, W. Chen, Q. Xia, B. Guo, Q. Wang, S. Liu, Y. Liu, J. Li, H. Yu, Efficient cleavage of lignin-carbohydrate complexes and ultrafast extraction of lignin oligomers from wood biomass by microwave-assisted treatment with deep eutectic solvent, ChemSusChem 10 (8) (2017) 1692–1700, https://doi.org/10.1002/cssc.201601795.
[2] S. Wang, G. Dai, H. Yang, Z. Luo, Lignocellulosic biomass pyrolysis mechanism: a state-of-the-art review, Prog. Energ. 62 (2017) 33–86, https://doi.org/10.1016/j.pecs.2017.05.004.

[3] S.H. Lee, T.V. Doherty, R.J. Linhardt, J.S. Dordick, Ionic liquid-mediated selective extraction of lignin from wood leading to enhanced enzymatic cellulose hydrolysis, Biotechnol. Bioeng. 102 (5) (2009) 1368–1376, https://doi.org/10.1002/bit.22179.
[4] G. Bali, X. Meng, J.I. Deneff, Q. Sun, A.J. Ragauskas, The effect of alkaline pretreatment methods on cellulose structure and accessibility, ChemSusChem 8 (2) (2015) 275–279, https://doi.org/10.1002/cssc.201402752.
[5] R. Sankaran, R.A.P. Cruz, H. Pakalapati, P.L. Show, T.C. Ling, W.H. Chen, Y. Tao, Recent advances in the pretreatment of microalgal and lignocellulosic biomass: a comprehensive review, Bioresour. Technol. 298 (2020), 122476, https://doi.org/10.1016/j.biortech.2019.122476.
[6] T.H. Kim, H.J. Ryu, K.K. Oh, Improvement of organosolv fractionation performance for rice husk through a low acid-catalyzation, Energies 12 (9) (2019) 1800, https://doi.org/10.3390/en12091800.
[7] S. Ventura-Cruz, N. Flores-Alamo, A. Tecante, Preparation of microcrystalline cellulose from residual rose stems (Rosa spp.) by successive delignification with alkaline hydrogen peroxide, Int. J. Biol. Macromol. 155 (2020) 324–329, https://doi.org/10.1016/j.ijbiomac.2020.03.222.
[8] Y. Oh, S. Park, E. Yoo, S. Jo, J. Hong, H.J. Kim, K.J. Kim, K.K. Oh, S.H. Lee, Dihydrogen-bonding deep eutectic solvents as reaction media for lipase-catalyzed transesterification, Biochem. Eng. J. 142 (2019) 34–40, https://doi.org/10.1016/j.bej.2018.11.010.
[9] S. Khandelwal, Y.K. Tailor, M. Kumar, Deep eutectic solvents (DESs) as eco-friendly and sustainable solvent/catalyst systems in organic transformations, J. Mol. Liq. 215 (2016) 345–386, https://doi.org/10.1016/j.molliq.2015.12.015.
[10] A.P. Abbott, D. Boothby, G. Capper, D.L. Davies, R.K. Rasheed, Deep eutectic solvents formed between choline chloride and carboxylic acids: versatile alternatives to ionic liquids, J. Am. Chem. Soc. 126 (29) (2004) 9142–9147, https://doi.org/10.1021/ja048266j.
[11] Y.P. Mbous, M. Hayyan, A. Hayyan, W.F. Wong, M.A. Hashim, C.Y. Looi, Applications of deep eutectic solvents in biotechnology and bioengineering-promises and challenges, Biotechnol. Adv. 35 (2) (2017) 105–134, https://doi.org/10.1016/j.biotechadv.2016.11.006.
[12] M. Francisco, A. van den Bruinhorst, M.C. Kroon, New natural and renewable low transition temperature mixtures (LTTMs): screening as solvents for lignocellulosic biomass processing, Green Chem. 14 (8) (2012) 2153–2157, https://doi.org/10.1039/C2GC35660K.
[13] C. Alvarez-Vasco, R. Ma, M. Quintero, M. Guo, S. Geleynse, K.K. Ramasamy, M. Wolcott, X. Zhang, Unique low-molecular-weight lignin with high purity extracted from wood by deep eutectic solvents (DES): a source of lignin for valorization, Green Chem. 18 (19) (2016) 5133–5141, https://doi.org/10.1039/C6GC01007E.
[14] W. Xing, G. Xu, J. Dong, R. Han, Y. Ni, Novel dihydrogen-bonding deep eutectic solvents: pretreatment of rice straw for butanol fermentation featuring enzyme recycling and high solvent yield, Chem. Eng. J. 333 (2018) 712–720, https://doi.org/10.1016/j.cej.2017.09.176.
[15] Y.W. Sai, K.M. Lee, Enhanced cellulase accessibility using acid-based deep eutectic solvent in pretreatment of empty fruit bunches, Cellulose 26 (2019) 9517–9528, https://doi.org/10.1007/s10570-019-02770-w.
[16] Q. Liu, T. Yuan, Q.J. Fu, Y.Y. Bai, F. Peng, C.L. Yao, Choline chloride-lactic acid deep eutectic solvent for delignification and nanocellulose production of moso bamboo, Cellulose 26 (2019) 9447–9462, https://doi.org/10.1007/s10570-019-02726-0.
[17] J. Hu, W. Zhu, Q. Yang, G. Qian, H. Xing, Solvatochromic parameters of the binary mixtures of imidazolium chloride ionic liquid plus molecular solvent, J. Appl. Sol. Chem. 3 (2014) 223–230, https://doi.org/10.6000/1929-5030.2014.03.04.4.
[18] P.G. Jessop, D.A. Jessop, D. Fu, L. Phan, Solvatochromic parameters for solvents of interest in green chemistry, Green Chem. 14 (5) (2012) 1245–1259, https://doi.org/10.1039/C2GC16670D.
[19] R. Pardo, M. Zayat, D. Levy, $E_T(33)$ dye as a tool for polarity determinations: application to porous hybrid silica thin-films, J. Photoch. Photobio. A. 210 (1) (2010) 17–22, https://doi.org/10.1016/j.jphotochem.2009.12.011.
[20] T.J. Park, S.H. Lee, Deep eutectic solvent systems for $FeCl_3$-catalyzed oxidative polymerization of 3-octylthiophene, Green Chem. 19 (4) (2017) 910–913, https://doi.org/10.1039/C6GC02789J.
[21] H. Zhao, G.A. Baker, Z. Song, O. Olubajo, L. Zanders, S.M. Campbell, Effect of ionic liquid properties on lipase stabilization under microwave irradiation, J. Mol. Catal. B Enzym. 57 (1–4) (2009) 149–157, https://doi.org/10.1016/j.molcatb.2008.08.006.
[22] T.V. Doherty, M. Mora-Pale, S.E. Foley, R.J. Linhardt, J.S. Dordick, Ionic liquid solvent properties as predictors of lignocellulose pretreatment efficacy, Green Chem. 12 (11) (2010) 1967–1975, https://doi.org/10.1039/C0GC00206B.
[23] S.H. Kim, S. Park, H. Yu, J.H. Kim, H.J. Kim, Y.H. Yang, Y.H. Kim, K.J. Kim, E. Kan, S.H. Lee, Effect of deep eutectic solvent mixtures on lipase activity and stability, J. Mol. Catal. B Enzym. 128 (2016) 65–72, https://doi.org/10.1016/j.molcatb.2016.03.012.
[24] A. Sluiter, B. Hames, R. Ruiz, C. Scarlata, J. Sluiter, D. Templeton, D. Crocker, Determination of structural carbohydrates and lignin in biomass, Laboratory Analytical Procedure, National Renewable Energy Laboratory, 2008.
[25] G.L. Miller, Use of dinitrosalicylic acid reagent for determination of reducing sugar, Anal. Chem. 31 (3) (1959) 426–428, https://doi.org/10.1021/ac60147a030.
[26] X. Ju, M. Bowden, E.E. Brown, X. Zhang, An improved X-ray diffraction method for cellulose crystallinity measurement, Carbohydr. Polym. 123 (2015) 476–481, https://doi.org/10.1016/j.carbpol.2014.12.071.
[27] J.G. Lynam, M.T. Reza, V.R. Vasquez, C.J. Coronella, Pretreatment of rice hulls by ionic liquid dissolution, Bioresour. Technol. 114 (2012) 629–636, https://doi.org/10.1016/j.biortech.2012.03.004.
[28] J.G. Lynam, N. Kumar, M.J. Wong, Deep eutectic solvents' ability to solubilize lignin, cellulose, and hemicellulose; thermal stability; and density, Bioresour. Technol. 238 (2017) 684–689, https://doi.org/10.1016/j.biortech.2017.04.079.
[29] E.R.E. Hassan, F. Mutelet, J.C. Moïse, From the dissolution to the extraction of carbohydrates using ionic liquids, RSC Adv. 3 (43) (2013) 20219–20226, https://doi.org/10.1039/C3RA42640H.
[30] D.J.G.P. van Osch, L.J.B.M. Kollau, A. van den Bruinhorst, S. Asikainen, M.A.A. Rocha, M.C. Kroon, Ionic liquids and deep eutectic solvents for lignocellulosic biomass fractionation, Phys. Chem. Chem. Phys. 19 (4) (2017) 2636–2665, https://doi.org/10.1039/C6CP07499E.
[31] X. Shen, T. Chen, H. Wang, Q. Mei, F. Yue, S. Sun, J. Wen, T. Yuan, R. Sun, Structural and morphological transformations of lignin macromolecules during bio-based deep eutectic solvent (DES) pretreatment, ACS Sustain. Chem. Eng. 8 (2020) 2130–2137, https://doi.org/10.1021/acssuschemeng.9b05106.
[32] A.M. da Costa Lopes, J.R.B. Gomes, J.A.P. Coutinho, A.J.D. Silvestre, Novel insights into biomass delignification with acidic deep eutectic solvents: a mechanistic study of β-O-4 ether bond cleavage and the role of the halide counterion in the catalytic performance, Green Chem. 22 (8) (2020) 2474–2487, https://doi.org/10.1039/C9GC02569C.
[33] C. Fang, M.H. Thomsen, C.G. Frankær, G.P. Brudecki, J.E. Schmidt, I.M. AlNashef, Reviving pretreatment effectiveness of deep eutectic solvents on lignocellulosic date palm residues by prior recalcitrance reduction, Ind. Eng. Chem. Res. 56 (12) (2017) 3167–3174, https://doi.org/10.1021/acs.iecr.6b04733.
[34] A.I. Yeh, Y.C. Huang, S.H. Chen, Effect of particle size on the rate of enzymatic hydrolysis of cellulose, Carbohydr. Polym. 79 (1) (2010) 192–199, https://doi.org/10.1016/j.carbpol.2009.07.049.
[35] M. Eibinger, P. Bubner, T. Ganner, H. Plank, B. Nidetzky, Surface structural dynamics of enzymatic cellulose degradation, revealed by combined kinetic and atomic force microscopy studies, FEBS J. 281 (2014) 275–290, https://doi.org/10.1111/febs.12594.
[36] J. Gong, J. Li, J. Xu, Z. Xiang, L. Mo, Research on cellulose nanocrystals produced from cellulose sources with various polymorphs, RSC Adv. 7 (45) (2017) 3486–33493, https://doi.org/10.1039/c7ra06222b.
[37] Y.L. Loowa, T.Y. Wu, G.H. Yang, L.Y. Ang, E.K. New, L.F. Siow, J.M. Jahimd, A.W. Mohammad, W.H. Teoh, Deep eutectic solvent and inorganic salt pretreatment of lignocellulosic biomass for improving xylose recovery, Bioresour. Technol. 249 (2018) 818–825, https://doi.org/10.1016/j.biortech.2017.07.165.
[38] W. Guo, Y. Hou, S. Ren, S. Tian, W. Wu, Formation of deep eutectic solvents by phenols and choline chloride and their physical properties, J. Chem. Eng. Data 58 (4) (2013) 866–872, https://doi.org/10.1021/je300997v.
[39] J. Shi, D. Xing, J. Lia, FTIR studies of the changes in wood chemistry from wood forming tissue under inclined treatment, Energy Procedia 16 (2012) 758–762, https://doi.org/10.1016/j.egypro.2012.01.122.
[40] M. Traore, J. Kaal, A.M. Cortizas, Differentiation between pine woods according to species and growing location using FTIR-ATR, Wood Sci. Technol. 52 (2018) 487–504, https://doi.org/10.1007/s00226-017-0967-9.
[41] G. Zhou, G. Taylor, A. Polle, FTIR-ATR-based prediction and modelling of lignin and energy contents reveals independent intra-specific variation of these traits in bioenergy poplars, Plant Methods 7 (2011) 1–10, https://doi.org/10.1186/1746-4811-7-9.
[42] K. Shahbaz, F.S. Mjalli, G. Vakili-Nezhaad, I.M. AlNashef, A. Asadov, M.M. Farid, Thermogravimetric measurement of deep eutectic solvents vapor pressure, J. Mol. Liq. 222 (2016) 61–66, https://doi.org/10.1016/j.molliq.2016.06.106.